



REPRODUCIBILITY AND ANGULAR MISALIGNMENT STUDY FOR TRANSMISSION MEASUREMENTS OF BGO

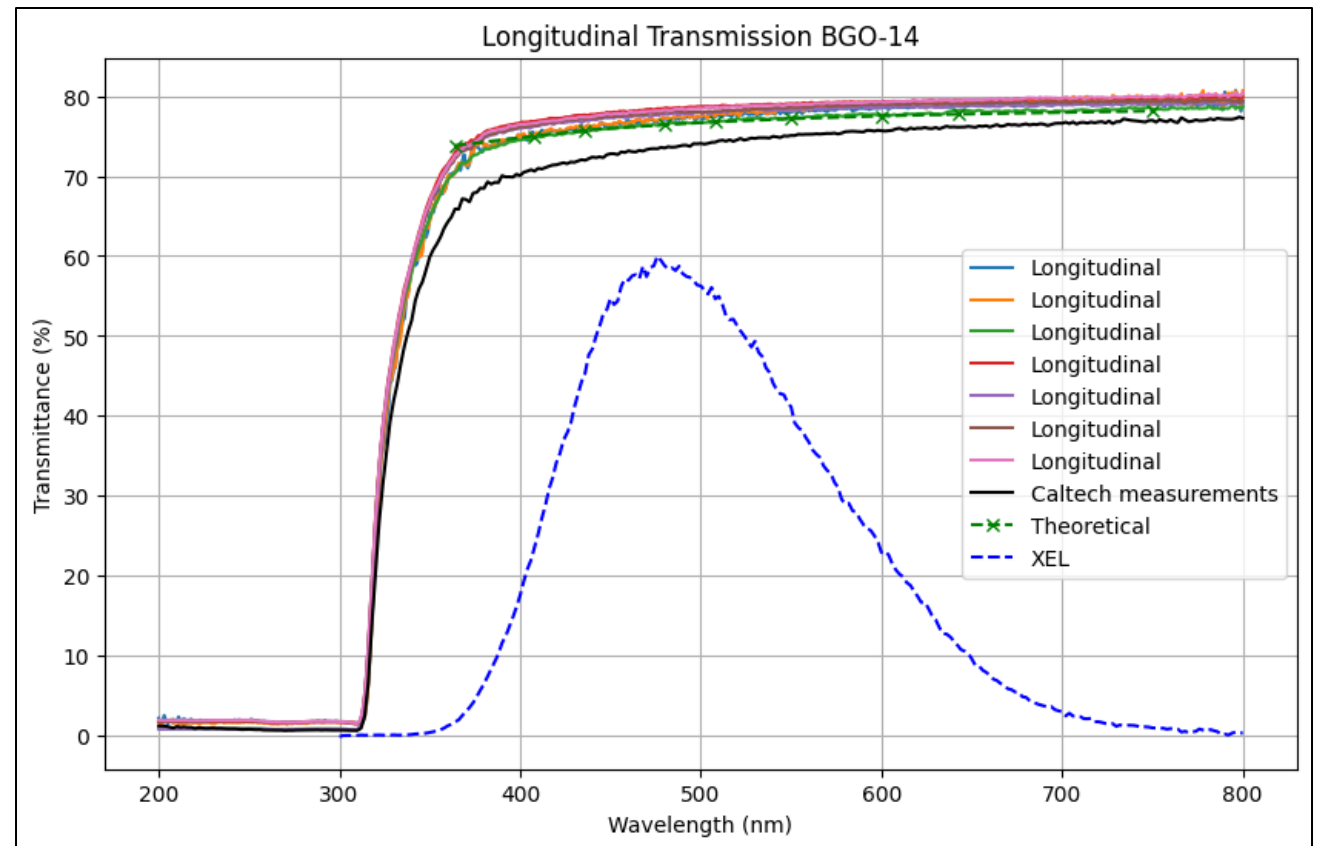
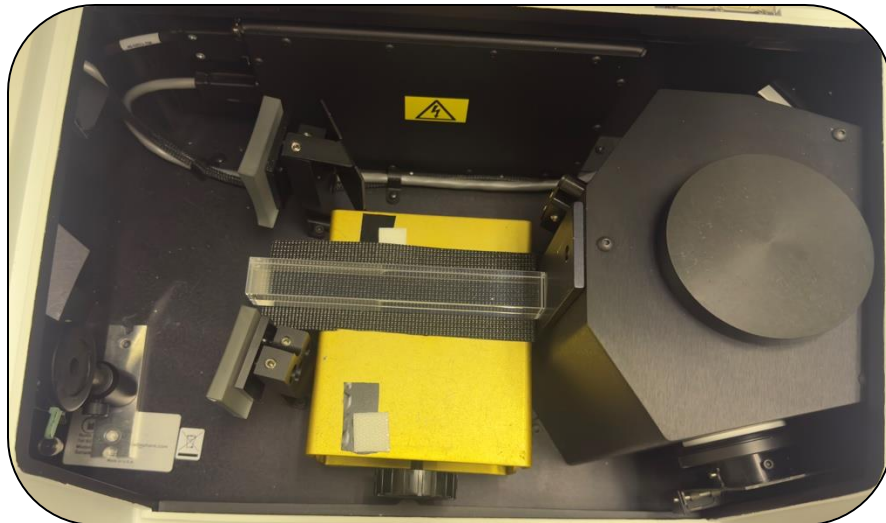
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Liyuan Zhang (CALTECH), Shin-Shan Eiko Yu

TRANSMISSION MEASUREMENTS FOR BGO-14

- We received two of the marked BGO crystals from UVA: BGO-7 and BGO-14.
- We used the “Perkin Elmer Lambda 950” Spectrophotometer to perform repeated transmission measurements on BGO-14 to check the reproducibility of results and to check for systematic uncertainties in measurements.
- We took repeated measurements by reseating the crystal in both longitudinal and transverse orientation and study the variability of transmittance.
- Results are also compared with the transmittance measurements made at CALTECH Crystal HEP lab.
- We also measured transmittance by introducing systematic angular misalignment with respect to the incoming beam to study the uniformity in the results.

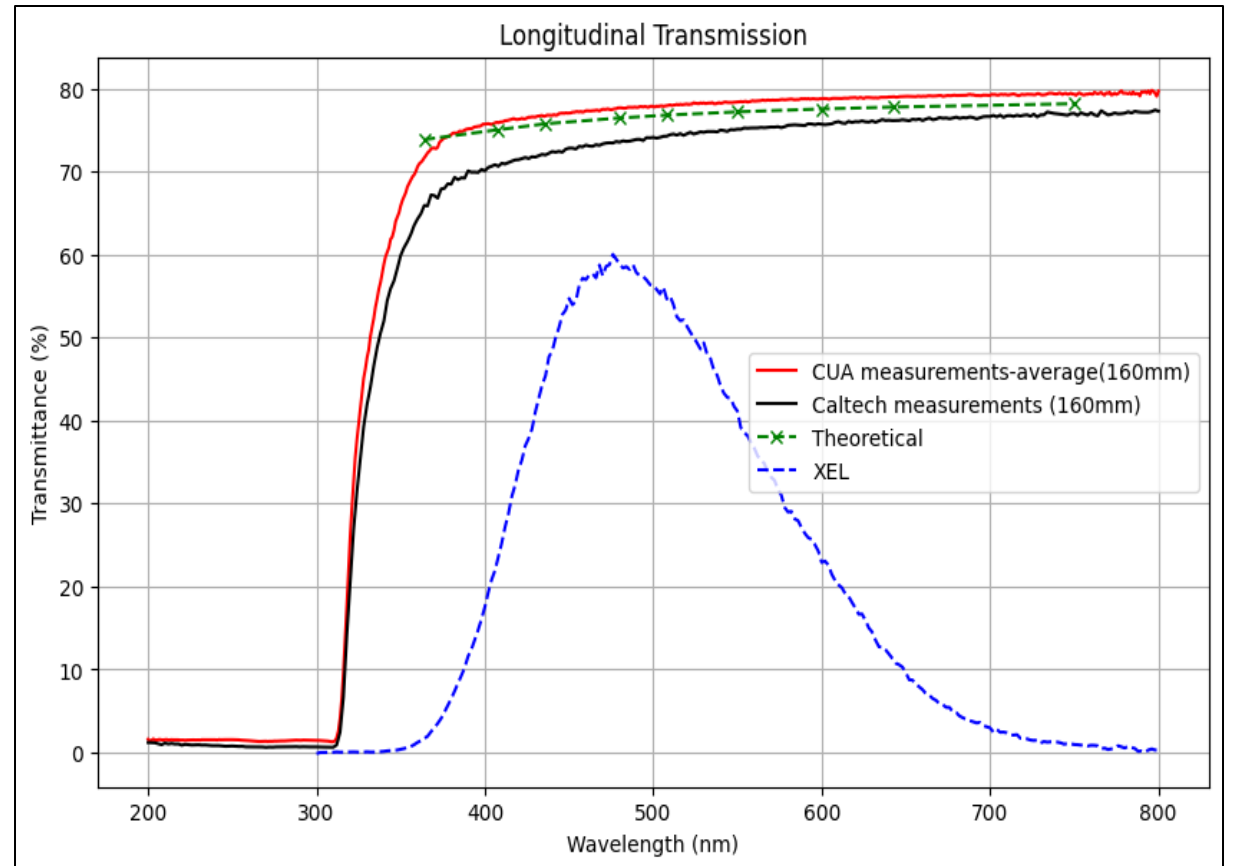
TRANSMISSION MEASUREMENTS FOR BGO-14

- Re-seating of crystal gives reproducible results in both Longitudinal and Transverse orientations.
- Standard deviation calculated for transmittance at 480nm. Std. Dev = 0.7%



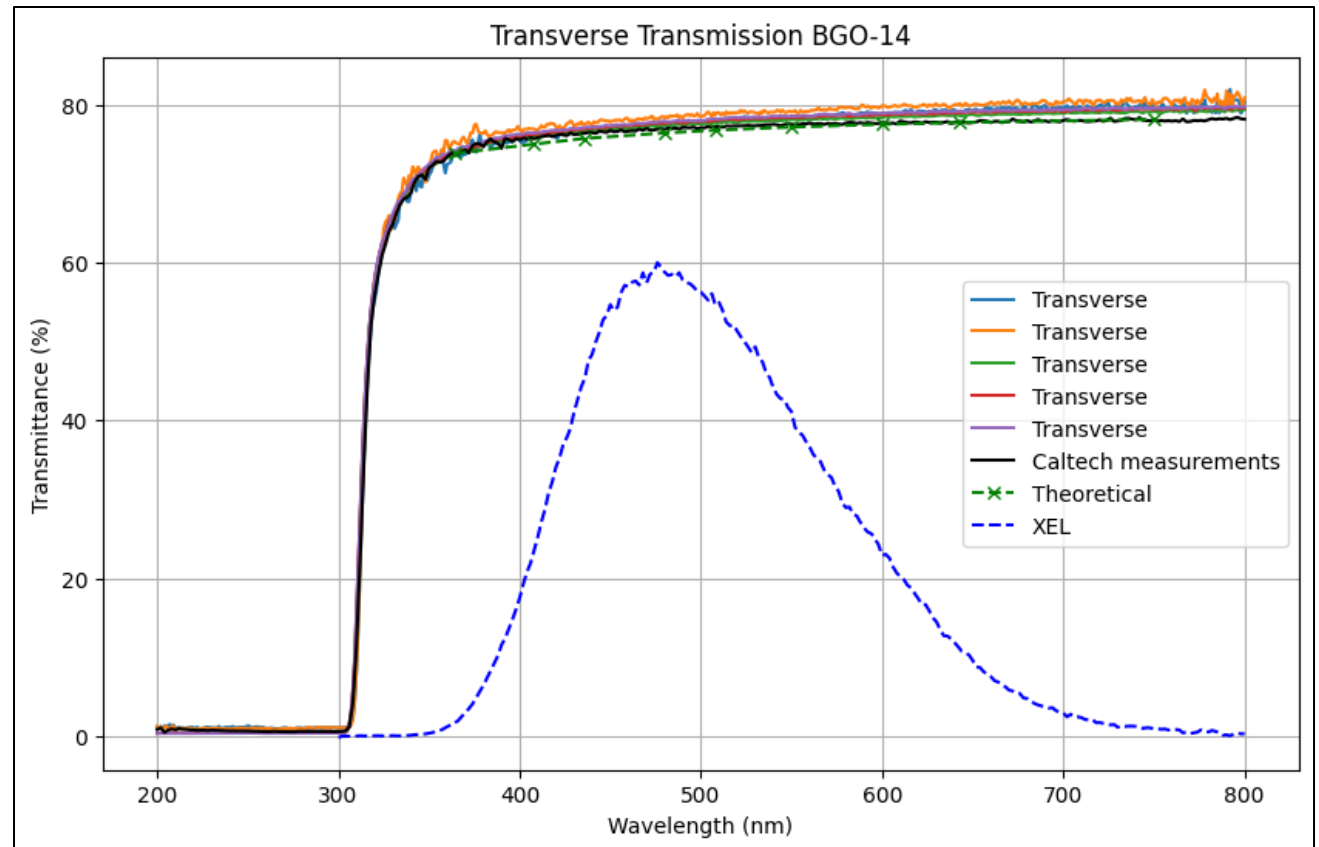
TRANSMISSION MEASUREMENTS FOR BGO-14

- Variation in the average longitudinal transmittance measured at CUA and transmittance measured at CALTECH can be observed
- Could be due to slightly different configuration settings or difference in the intensity of beam
- Emission weighted longitudinal transmission for the two measurements
 - $\text{EWLT_CUA} = 77.7\%$
 - $\text{EWLT_Caltech} = 73.9\%$



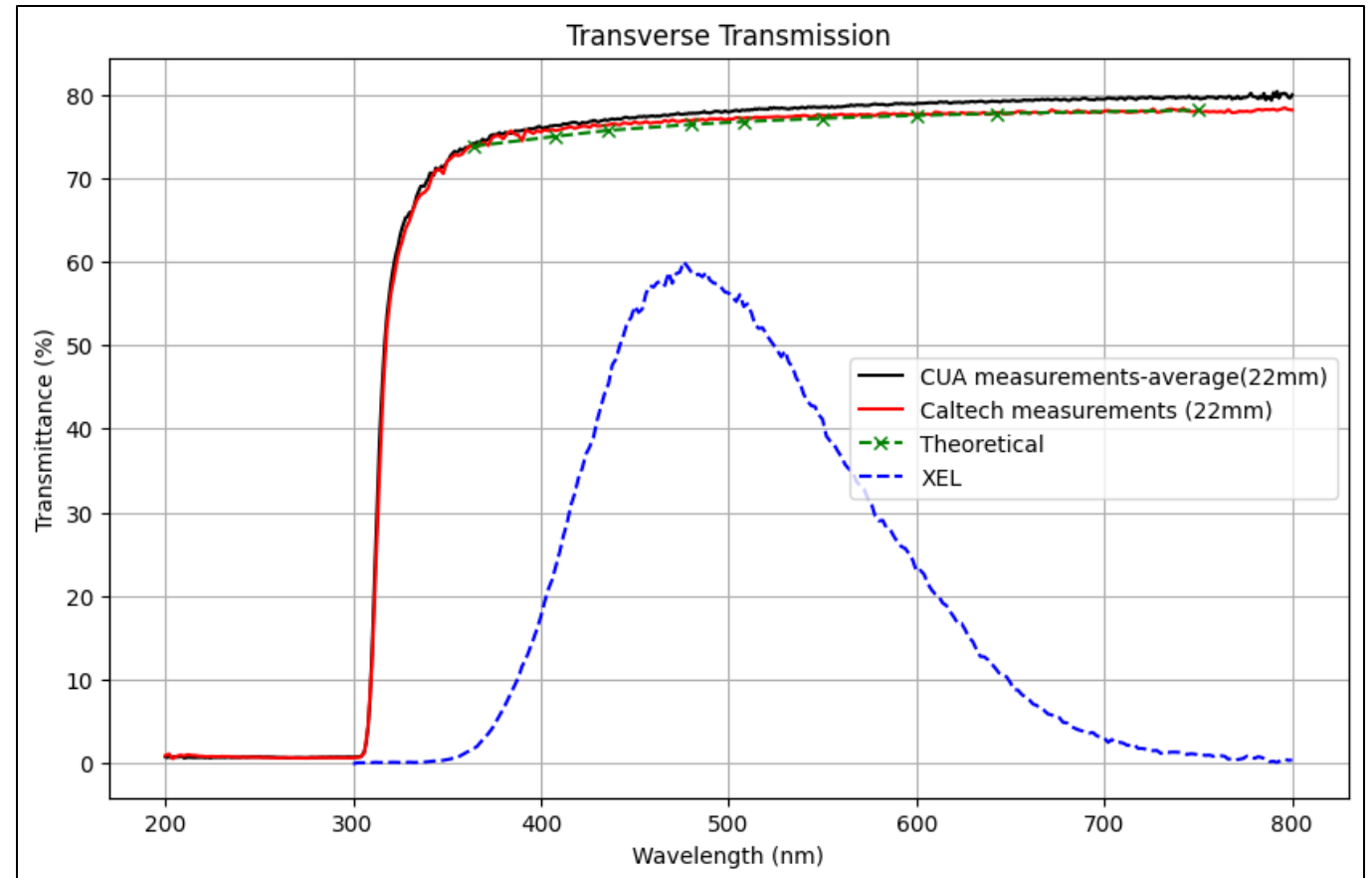
TRANSMISSION MEASUREMENTS FOR BGO-14

- Even better reproducibility was seen in transverse measurements.
- Standard deviation calculated for transmittance at 480nm. Std. Dev = 0.4%



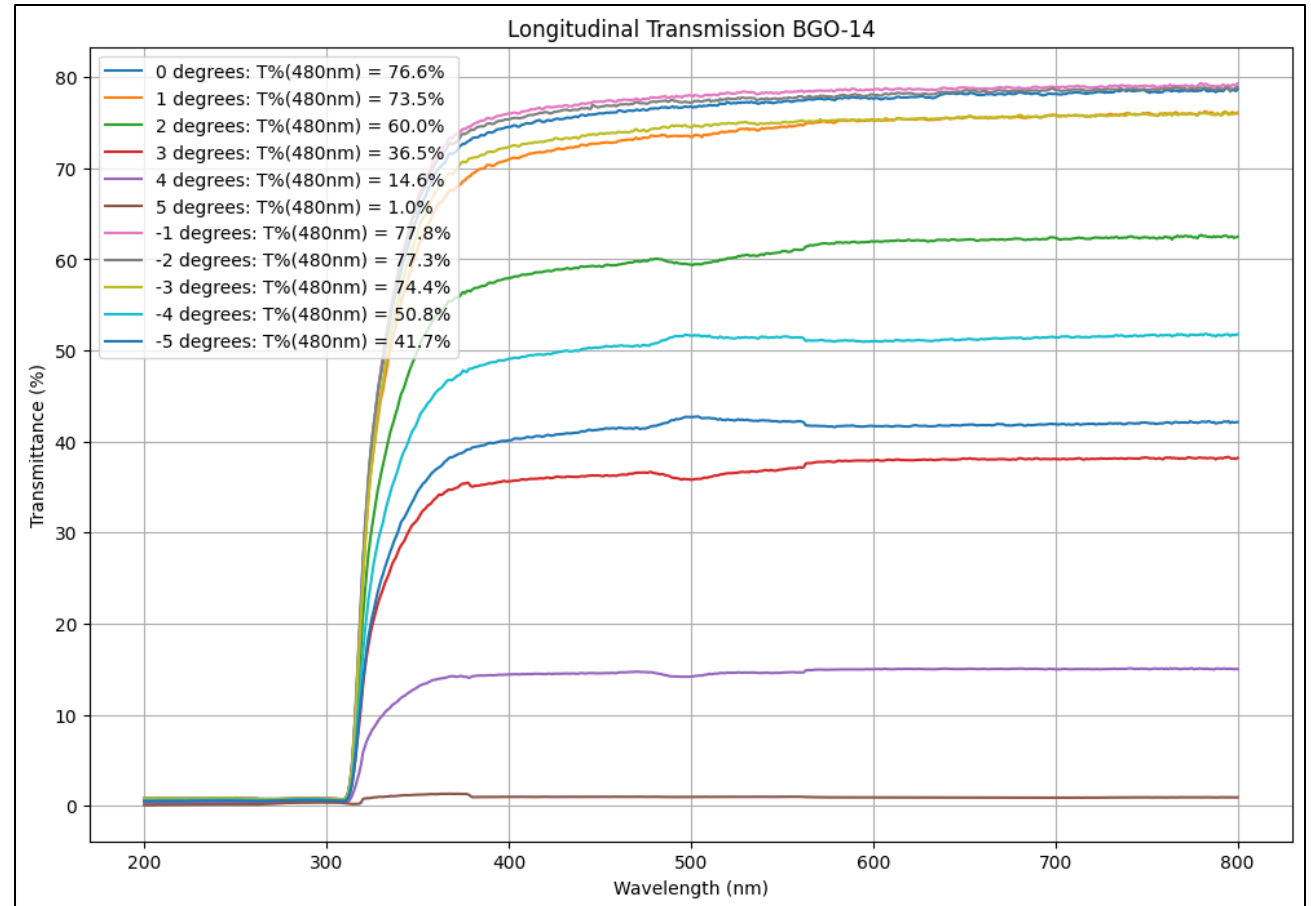
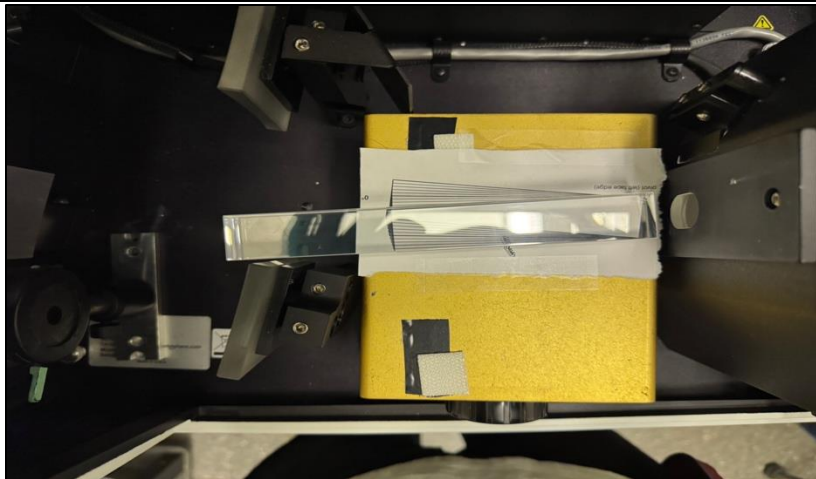
TRANSMISSION MEASUREMENTS FOR BGO-14

- Variation in the average transverse transmittance measured at CUA and transmittance measured at CALTECH is not significant.
- Emission weighted transverse transmission for the two measurements
 - EWTT_CUA = 77.9%
 - EWTT_Caltech = 77.0 %



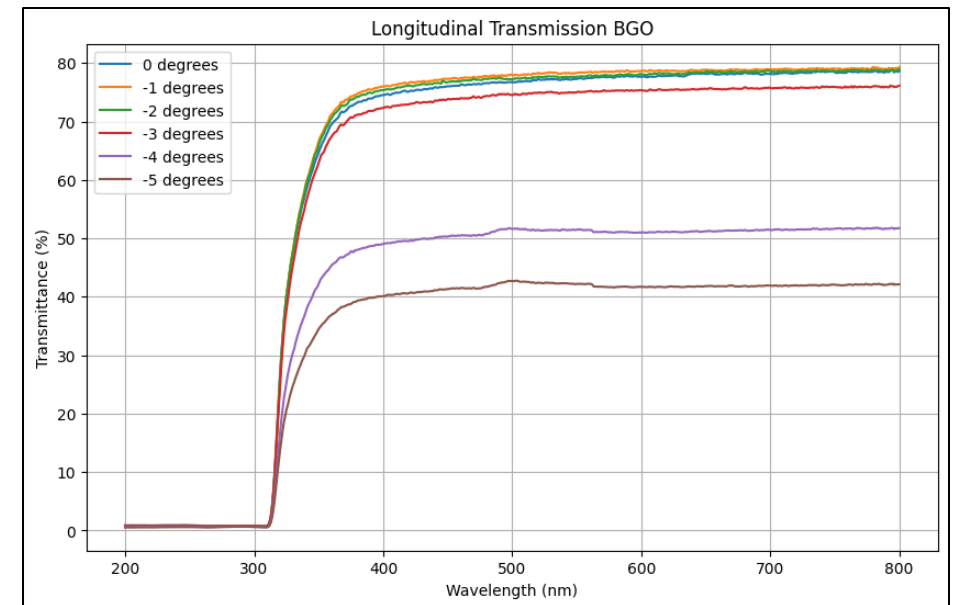
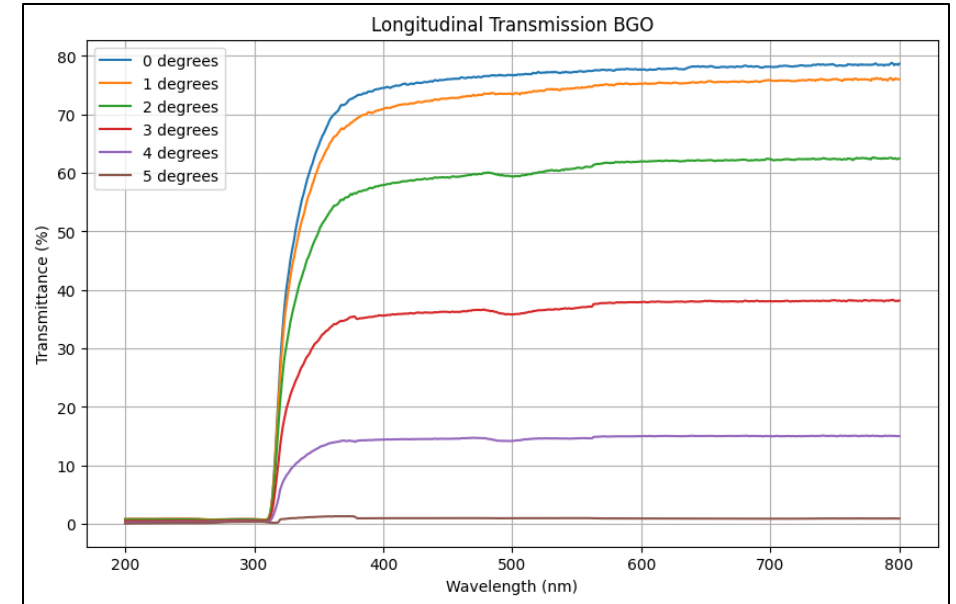
TRANSMISSION MEASUREMENTS FOR BGO-14

- Angular variations in longitudinal orientation.
- Angular guide was used to rotate the crystal along an axis passing through the end face of crystal close to the PMT.
- Angular variations were made from -5 to +5 degrees.



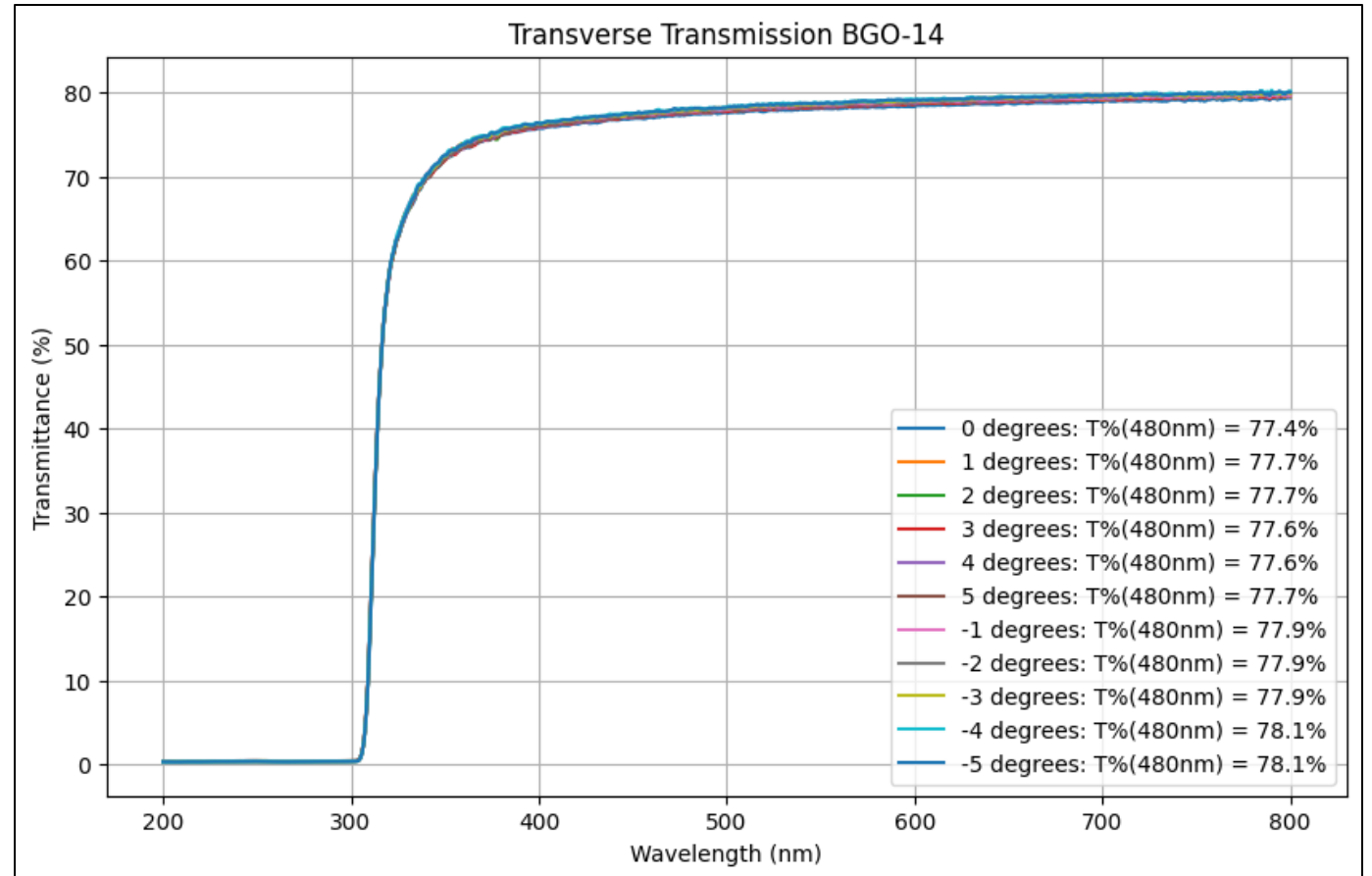
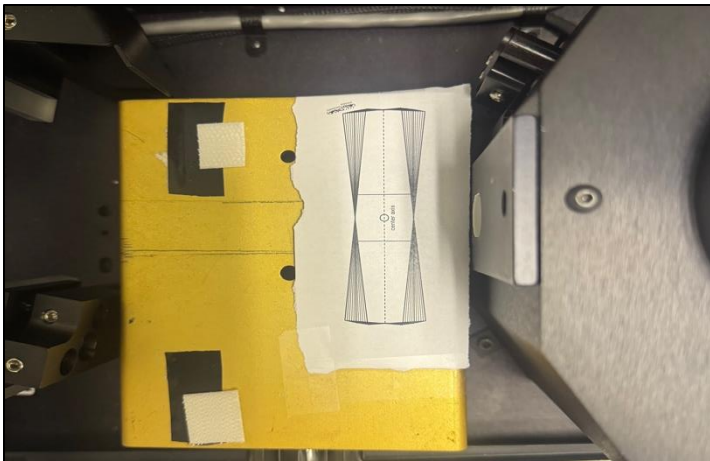
TRANSMISSION MEASUREMENTS FOR BGO-14

- The variation in transmittance was not uniform when measured in either directions.
- Possibly because of more internal reflection in one orientation than in symmetrically opposite orientation.
- Could also be due to unequal aperture acceptance of end face of crystal with respect to the PMT.
- Thoughts?



TRANSMISSION MEASUREMENTS FOR BGO-14

- Angular variations in Transverse orientation.
- Angular guide was used to rotate the crystal along an axis passing through the end face of crystal close to the PMT.
- Angular variations were made from -5 to +5 degrees.
- No significant variation was observed



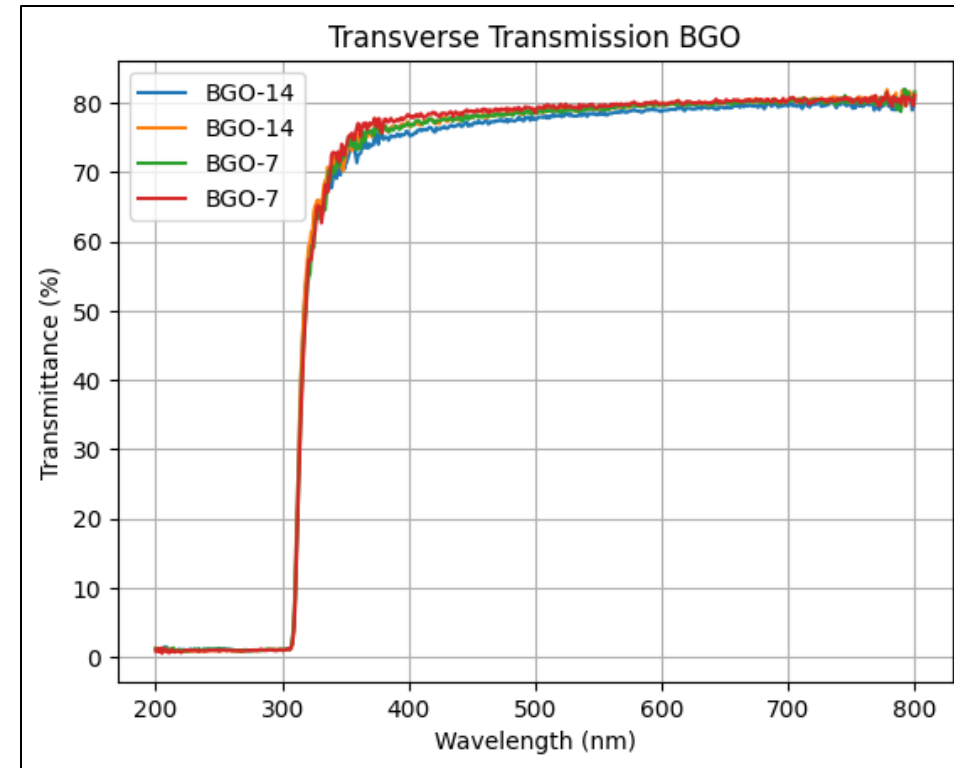
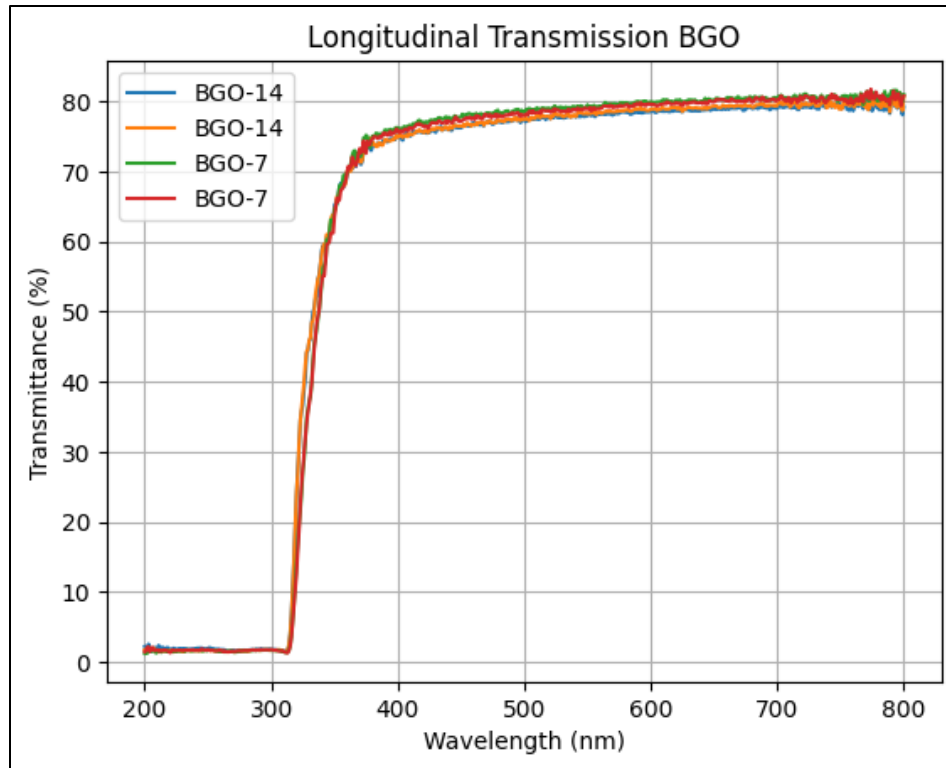
SUMMARY

- **Reproducibility:**
 - Excellent consistency in both longitudinal and transverse orientations.
 - Std. dev. at 480 nm: **0.7% (Longitudinal)**, **0.4% (Transverse)**.
- **CUA vs. Caltech Comparison:**
 - Small but noticeable difference in longitudinal EWLT values.
 - Transverse measurements largely consistent.
 - Possible causes: configuration differences, beam intensity variation.
- **Angular Misalignment Study:**
 - **Longitudinal:** Asymmetric variation observed; may stem from internal reflections or aperture acceptance differences.
 - **Transverse:** No significant angular dependence; highly stable.

THANK YOU! QUESTIONS OR THOUGHTS?

Special Thanks to Prof. Tanja Horn, CUA and her grad students Mr. Avnish Singh and Mr. Sameer Jain for assisting with these measurements.

BACKUP



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